

Group Assignment 3

Group 2

Yuxuan Niu
Kin Chung Shiu
Rui Ma
Yifan Deng

June 13, 2026

1. Consider a three state $(1, 2, 3)$ Markov chain with transition matrix

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0.5 \end{pmatrix}$$

- (a) Draw the transition diagram.
- (b) Find the stationary distribution π .
- (c) Given that the chain is in state 1 at time 1, what is the probability that the chain is in state 2 at time 4 ?
- (d) Given that the chain is in state 1 at time 1, what is the expected time until the chain is in state 3 the first time?
- (e) What is the period of each state?

Solution 1:

(a) The Markov chain has three states $\{1, 2, 3\}$ with the following transitions:

- From state 1: transitions to state 1 with probability 0.5 and to state 2 with probability 0.5.
- From state 2: transitions to state 1 with probability 0.5 and to state 3 with probability 0.5.
- From state 3: transitions to state 1 with probability 0.5 and to state 3 with probability 0.5.

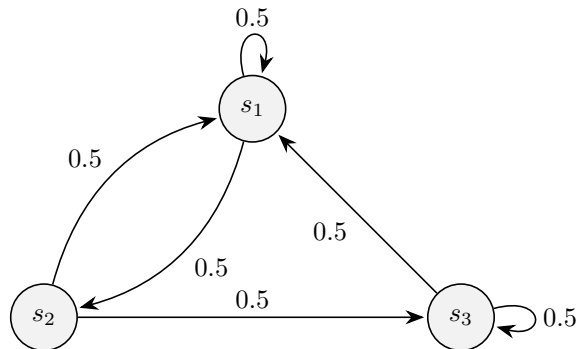


Figure 1: Transition Diagram

(b) The stationary distribution $\pi = (\pi_1, \pi_2, \pi_3)$ satisfies

$$\pi = \pi P, \quad \pi_1 + \pi_2 + \pi_3 = 1.$$

Solving the system yields

$$\pi = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right).$$

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(d) Let h_i denote the expected time to first reach state 3 starting from state i . We have $h_3 = 0$ and

$$\begin{aligned} h_1 &= 1 + 0.5h_1 + 0.5h_2, \\ h_2 &= 1 + 0.5h_1. \end{aligned}$$

Solving the system gives $h_1 = 6$. (e) State 1 and state 3 have self-loops, hence their periods are 1. State 2 can return to itself in both even and odd numbers of steps. Therefore, all states have period 1.

2. Assume that we are trying to classify a binary outcome Y , i.e. our data is of the form $(X, Y) \sim F_{X,Y}$, where $Y \in \{0, 1\}$ and $X \in \mathbb{R}^d$. We have used data to train a classifier $g(X)$. We can evaluate the performance of the classifier using i.i.d. testing data, $(X_1, Y_1), \dots, (X_n, Y_n)$. We are interested in estimating the following quantities:

Precision : $\mathbb{P}(Y = 1 \mid g(X) = 1)$,

Recall : $\mathbb{P}(g(X) = 1 \mid Y = 1)$.

(a) Write down the empirical version of the precision and recall.

(b) Let us now think that the variable Y denotes if a battery's health has deteriorated or not, and let X denote a bunch of constructed health indicators about the battery. If the model $g(X)$ predicts that the battery has deteriorated you need to run a test to confirm this. The cost of running the test is c when the battery is not deteriorated. On the other hand, if the battery is in fact deteriorated and the test is not run, the battery will die during use and the cost of this is d . Define a random variable representing the cost of the decision $g(X)$ and write down the formula for the expected cost in terms of the precision and recall.

(c) Advanced question: can you produce a confidence interval for the expected cost? What about the precision and the recall?

Solution :2

(a) Empirical Version of Precision and Recall.

Let $\mathbb{I}(\cdot)$ be the indicator function, which equals 1 if the condition is true and 0 otherwise. Given the i.i.d. testing data $(X_1, Y_1), \dots, (X_n, Y_n)$:

• Empirical Precision:

$$\widehat{\text{Precision}} = \frac{\sum_{i=1}^n \mathbb{I}(g(X_i) = 1 \text{ and } Y_i = 1)}{\sum_{i=1}^n \mathbb{I}(g(X_i) = 1)}$$

• Empirical Recall:

$$\widehat{\text{Recall}} = \frac{\sum_{i=1}^n \mathbb{I}(g(X_i) = 1 \text{ and } Y_i = 1)}{\sum_{i=1}^n \mathbb{I}(Y_i = 1)}$$

(b) Expected Cost in terms of Precision and Recall.

Let C be the random variable representing the cost of the decision $g(X)$. Based on the problem description:

- Cost of False Positive (Type I error): c (Test cost when battery is healthy).
- Cost of False Negative (Type II error): d (Damage cost when battery deteriorates).
- Cost of Correct Decision: 0.

The cost function is defined as:

$$C = \begin{cases} c, & \text{if } g(X) = 1 \text{ and } Y = 0 \\ d, & \text{if } g(X) = 0 \text{ and } Y = 1 \\ 0, & \text{otherwise} \end{cases}$$

The Expected Cost is:

$$\mathbb{E}[C] = c \cdot \mathbb{P}(g(X) = 1, Y = 0) + d \cdot \mathbb{P}(g(X) = 0, Y = 1)$$

Let $p = \mathbb{P}(Y = 1)$ be the prior probability of battery deterioration. Let π denote Precision and ρ denote Recall.

1. Analyzing False Negatives: By the definition of Recall $\rho = \mathbb{P}(g(X) = 1 \mid Y = 1)$:

$$\mathbb{P}(g(X) = 1, Y = 1) = \rho p$$

Thus, the False Negative probability is:

$$\mathbb{P}(g(X) = 0, Y = 1) = \mathbb{P}(Y = 1) - \mathbb{P}(g(X) = 1, Y = 1) = p(1 - \rho)$$

2. Analyzing False Positives: By the definition of Precision $\pi = \frac{\mathbb{P}(Y=1, g(X)=1)}{\mathbb{P}(g(X)=1)}$:

$$\mathbb{P}(g(X) = 1) = \frac{\mathbb{P}(Y = 1, g(X) = 1)}{\pi} = \frac{\rho p}{\pi}$$

The False Positive probability is:

$$\begin{aligned} \mathbb{P}(g(X) = 1, Y = 0) &= \mathbb{P}(g(X) = 1) - \mathbb{P}(g(X) = 1, Y = 1) \\ &= \frac{\rho p}{\pi} - \rho p \\ &= \rho p \left(\frac{1}{\pi} - 1 \right) = \rho p \left(\frac{1 - \pi}{\pi} \right) \end{aligned}$$

Final Formula: Substituting these back into the expectation equation:

$$\boxed{\mathbb{E}[C] = p \left[c \cdot \rho \left(\frac{1 - \pi}{\pi} \right) + d \cdot (1 - \rho) \right]}$$

(c) Advanced Question: Confidence Intervals. Yes. The cost C_i for each data point is an i.i.d. random variable taking values in $\{0, c, d\}$. By the Central Limit Theorem (CLT), the sample mean cost $\bar{C}$ is approximately normally distributed for large n . An approximate 95% confidence interval is:

$$\bar{C} \pm 1.96 \frac{\hat{\sigma}_C}{\sqrt{n}}$$

where $\hat{\sigma}_C$ is the sample standard deviation of the costs.

Precision and Recall are estimates of population proportions (specifically conditional probabilities).

- Since the denominator for Precision (number of predicted positives) is random, the standard binomial interval is an approximation.
- A robust method is to use **Bootstrapping** to estimate the variance and construct confidence intervals for both metrics.

3. Let X and Y be two d -dimensional zero mean, unit variance Gaussian random vectors. Show that X and Y are nearly orthogonal by calculating their dot product. Can you for instance also bound the probability that the dot product is larger than ε ?

Solution 3:

(a) Showing X and Y are nearly orthogonal.

Let $X = (X_1, \dots, X_d)^T$ and $Y = (Y_1, \dots, Y_d)^T$. Since X and Y are zero mean, unit variance Gaussian random vectors, we assume their components are independent and identically distributed (i.i.d.) with $X_i, Y_i \sim \mathcal{N}(0, 1)$.

The dot product is given by:

$$Z = X \cdot Y = \sum_{i=1}^d X_i Y_i$$

Calculating the Expectation: Since X_i and Y_i are independent:

$$\mathbb{E}[Z] = \sum_{i=1}^d \mathbb{E}[X_i Y_i] = \sum_{i=1}^d \mathbb{E}[X_i] \mathbb{E}[Y_i] = \sum_{i=1}^d 0 \cdot 0 = 0$$

This shows that the vectors are orthogonal *on average*.

Calculating the Variance: The variance of the product of two independent standard normal variables is:

$$\text{Var}(X_i Y_i) = \mathbb{E}[X_i^2 Y_i^2] - (\mathbb{E}[X_i Y_i])^2 = \mathbb{E}[X_i^2] \mathbb{E}[Y_i^2] - 0 = 1 \cdot 1 = 1$$

Since the dimensions are independent:

$$\text{Var}(Z) = \sum_{i=1}^d \text{Var}(X_i Y_i) = \sum_{i=1}^d 1 = d$$

Thus, the standard deviation of the dot product is $\sigma_Z = \sqrt{d}$.

Conclusion on Orthogonality: To define "nearly orthogonal", we compare the dot product to the magnitude of the vectors. By the Law of Large Numbers, the squared Euclidean norm $\|X\|^2 = \sum X_i^2 \approx d$, so $\|X\| \approx \sqrt{d}$ (and similarly for Y). The cosine of the angle θ between X and Y is:

$$\cos \theta = \frac{X \cdot Y}{\|X\| \|Y\|} \approx \frac{Z}{d}$$

Since Z has standard deviation $\sqrt{d}$, the term $\frac{Z}{d}$ has standard deviation $\frac{\sqrt{d}}{d} = \frac{1}{\sqrt{d}}$. As $d \rightarrow \infty$, $\cos \theta \rightarrow 0$, meaning the vectors become orthogonal.

(b) Bounding the Probability.

We want to bound $\mathbb{P}(X \cdot Y > \epsilon)$.

Using the mean and variance derived above:

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq \epsilon) \leq \frac{\text{Var}(Z)}{\epsilon^2}$$

Substituting the values:

$$\mathbb{P}(|X \cdot Y| \geq \epsilon) \leq \frac{d}{\epsilon^2}$$

This provides a polynomial bound on the probability.

4. Let $u_1, \dots, u_r$ be $n \times 1$ unit length vectors that are linearly independent, i.e. $\sum_{i=1}^r \alpha_i u_i = 0$ implies that $\alpha_i = 0$ for all i .

(a) Verify that the matrix $u_i u_i^T$ is a rank one matrix for all i . What is the null-space and range of $u_i u_i^T$?

(b) Verify that the matrix $U = \sum_{i=1}^r u_i u_i^T$ is a rank r matrix.

(c) i. If we perform SVD on U , are the vectors $u_1, \dots, u_r$ the same as the right singular vectors? If not, can you give an example?

ii. What if the vectors $u_1, \dots, u_r$ are all orthogonal? In this case, what are the singular values of U ?

Solution 4:

Let $u_1, \dots, u_r \in \mathbb{R}^n$ be unit-length and linearly independent vectors, and define

$$U = \sum_{i=1}^r u_i u_i^T.$$

(a)

For any $x \in \mathbb{R}^n$,

$$(u_i u_i^T)x = u_i(u_i^T x),$$

which is a scalar multiple of u_i . Hence,

$$\text{rank}(u_i u_i^T) = 1.$$

The range is

$$\mathcal{R}(u_i u_i^T) = \text{span}\{u_i\},$$

and the null space is

$$\mathcal{N}(u_i u_i^T) = \{x \in \mathbb{R}^n \mid u_i^T x = 0\}.$$

(b)

For any $x \in \mathbb{R}^n$,

$$Ux = \sum_{i=1}^r u_i(u_i^T x),$$

so

$$\mathcal{R}(U) \subseteq \text{span}\{u_1, \dots, u_r\}.$$

Since $u_1, \dots, u_r$ are linearly independent, this span has dimension r , and therefore

$$\text{rank}(U) = r.$$

(c. i)

In general, $u_1, \dots, u_r$ are not the right singular vectors of U , since

$$U^2 = \sum_{i,j} u_i(u_i^T u_j)u_j^T$$

contains cross terms when the vectors are not orthogonal.

Example: Take $n = 2, r = 2$. Let

$$u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad u_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

They are linearly independent but not orthogonal.

Then

$$U = u_1 u_1^T + u_2 u_2^T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}.$$

The eigenvectors of U (right singular vectors) are not u_1 or u_2 . Thus, u_1, u_2 are not the right singular vectors of U .

(c. ii)

If $u_1, \dots, u_r$ are orthonormal, then

$$Uu_j = u_j \quad (j = 1, \dots, r),$$

and any vector orthogonal to their span is mapped to zero.

Since U is symmetric and positive semidefinite, its singular values equal its eigenvalues. Thus,

$$\sigma_1 = \dots = \sigma_r = 1,$$

and the remaining singular values are zero.

5. Let $X \sim \text{Uniform}(B_1)$ and define $Y = \|X\|_2$ (the Euclidean norm).

(a) Find the distribution function of Y .

(b) What is the distribution of $\ln(1/Y)$?

(c) Calculate $\mathbb{E}[\ln(1/Y)]$, first by using the distribution function of Y and then by using the distribution function of $\ln(1/Y)$.

Solution 5:

- (a) In this question, X is a random vector uniformly distributed on the unit ball with dimension $= d$. The volume of a dimension ball of radius $= r$ is proportional to r^d .

Y is a random variable defined as $\|X\|_2$, which is the distance of the point from the origin.

The cumulative distribution function (CDF), $F_Y(y)$, is the probability that the point falls within a ball of radius y :

$$F_Y(y) = P(Y \leq y) = P(\|X\|_2 \leq y)$$

Since X is uniformly distributed, the probability is the ratio of the volume of the ball of radius y to the total volume of the unit ball:

$$F_Y(y) = \frac{\text{Vol}(B_y)}{\text{Vol}(B_1)} \approx \frac{y^d}{1^d} = y^d$$

Therefore, the distribution function of Y is

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ y^d & 0 \leq y \leq 1 \\ 1 & y > 1 \end{cases}$$

- (b) Let $Z = \ln(1/Y) = -\ln(Y)$.

First, find the range of Z . Since $0 < Y \leq 1$, we have $0 \leq Z < \infty$. We can find the distribution by looking at the CDF of Z , denoted $F_Z(z)$:

$$F_Z(z) = P(Z \leq z) = P(-\ln(Y) \leq z)$$

Rearranging the inequality:

$$P(\ln(Y) \geq -z)$$

Exponentiating both sides:

$$P(Y \geq e^{-z})$$

Using the CDF of Y found in part (a):

$$P(Y \geq e^{-z}) = 1 - P(Y \leq e^{-z}) = 1 - F_Y(e^{-z})$$

Substitute $F_Y(y) = y^d$:

$$F_Z(z) = 1 - (e^{-z})^d = 1 - e^{-dz}$$

This is the cumulative distribution function of an Exponential distribution with rate parameter $\lambda = d$.

- (c) From distribution of Y :

$$\begin{aligned}
E[\ln(1/Y)] &= \int_0^1 \ln(1/y) \cdot f_Y(y) \, dy \\
&= \int_0^1 \ln(1/y) (d \cdot y^{d-1}) \, dy \\
&= \int_0^1 \ln(1/y) \, dy^d \\
&= [\ln(1/y) \cdot y^d]_0^1 - \int_0^1 y^d d \ln(1/y) \\
&= 0 - \int_0^1 y^d \cdot \frac{1}{1/y} \cdot (-y^{-2}) dy \\
&= \int_0^1 y^{d-1} dy \\
&= \left[\frac{y^d}{d} \right]_0^1 \\
&= \frac{1}{d}
\end{aligned}$$

From distribution of $\ln 1/Y$:

From part (b), we know that $Z = \ln(1/Y)$ follows an Exponential distribution with rate $\lambda = d$. The expected value of an Exponential distribution with rate λ is simply $1/\lambda$.

$$\mathbb{E}[\ln(1/Y)] = \frac{1}{d}$$